SOFTWARE TESTING · FINAL PROJECT

Grammar-Based Fuzzing and Differential Testing of Small Hugging Face Chat Models

What it finds, and where its oracles fail

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AI-generated content (Claude). Review before use.

Chat models are components, but hard to test

They rarely crash

A crash oracle, the engine of most fuzzers, reveals almost nothing.

Input is natural language

No parser whose structure a fuzzer could exploit.

No single right answer

The test-oracle problem at its most acute.

Systems under test: three small open chat models from the Hugging Face Hub — Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct, TinyLlama-1.1B-Chat — run locally with `transformers`.

The hosted HF API client is implemented but was not used (no access token).

Is grammar fuzzing + differential testing effective — and what limits it?

RQ1 · Effectiveness

How many failures, of which kinds, and at what cost?

RQ2 · Oracles

What do differential and metamorphic oracles add beyond a known answer?

RQ3 · Guidance

Does coverage-guided expansion beat random expansion?

RQ4 · Oracle reliability

How often do the automatic oracles misjudge a reply?

Motivation: existing LLM fuzzers target jailbreaks, and metamorphic frameworks compare a model only with itself. Nobody measured how often these oracles are right.

Four ideas, one test case

Technique	What it does here	Precedent
Grammar-based fuzzing	A context-free grammar derives the wording of each request	LangFuzz, Fuzzing Book
Coverage guidance	Prefers grammar expansions not yet used (b = 0.8)	1-path coverage
Differential testing	Same prompt to 3 models; disagreement is a signal	Csmith, DeepXplore
Metamorphic testing	Paraphrases of one task must get the same answer	CheckList
Ground truth	**Each derivation carries its expected answer**	**our addition**

Verifiable tasks: arithmetic, letter counting, string reversal, JSON output, verbatim echo of hard Unicode — plus free-form `edge` requests with no expected answer.

chatfuzz: from skeleton to finding

Skeleton
count, ‘r’, “strawberry” → 3

→

Grammar
16 nonterminals, 41 expansions

→

Paraphrases
up to 3 wordings

→

3 models
greedy, 256 tokens

Ground truth
reply vs. expected answer → canonical answer

Differential
canonical answers differ across models

Metamorphic
canonical answers differ across paraphrases

Python · `transformers` · errors recorded as data · truncation flagged · 19 unit tests

Four experiments on a 4-core CPU, no GPU

Experiment	Setup	Answers
Main campaign	60 skeletons → 170 prompts × 3 models = 510 replies (420 graded)	RQ1, RQ2
Ablation 1	30 seeds × 120 prompts, random (b = 0) vs. guided (b = 0.8), no models	RQ3
Ablation 2	3 seeds × 30 prompts × 2 models, both values of b	RQ3
Oracle validation	Stratified sample of 60 replies judged by content and by strict format	RQ4

Pinned model revisions and greedy decoding make every reply reproducible: the 27 repeated prompts got identical replies. Statistics: Wilson 95% intervals, exact McNemar tests.

64.5%

of 420 graded replies failed their ground-truth oracle (271 failures)

Pass rate (%)	Qwen	SmolLM2	TinyLlama
arith	33	27	10
count	37	30	30
reverse	0	3	0
json	100	100	0
echo	25	90	75
overall	**40**	**47**	**19**

Reversal solved **1 time in 90**. Six-digit operands: 9% correct vs. 44%. TinyLlama, the largest model, wraps JSON in prose. 25 of 150 model–task pairs answer some paraphrases right and others wrong. Cost: 54 CPU-minutes, ~5 failures per minute.

Differential finds almost everything and decides almost nothing

Differential

98.8%

recall · 73.9% precision

Flags every model in a disagreement, including the right one.

Metamorphic

88.0%

precision · 79.5% recall

Misses models that are consistently wrong.

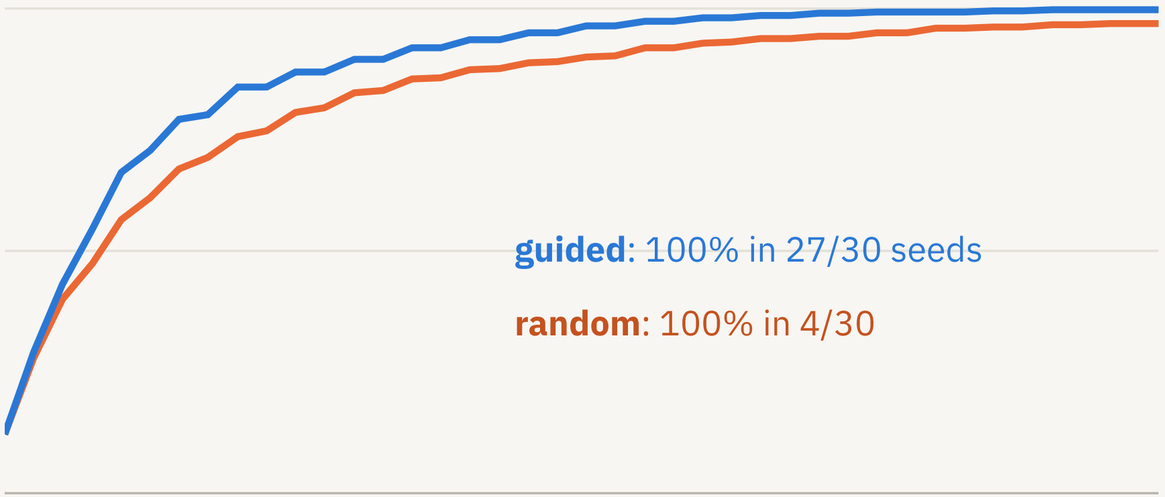
Who was right?

58/111

disagreements where no model was correct

Only 1 prompt got the same wrong answer from all three.

Guidance covers the grammar; failures barely change



Grammar coverage over 120 prompts, mean of 30 seeds. With models: 18.7 vs. 17.0 distinct failure signatures, 30.3 vs. 30.0 failures (3 seeds).

Oracles vs. manual judgment

95%

agreement on 60 replies (κ = 0.89)

All 3 errors: valid JSON wrapped in prose, rejected by the strict JSON oracle. A lenient oracle moves TinyLlama’s JSON score from **0% to 83%**.

Same task, three wordings, three answers

Prompt to Qwen2.5-0.5B (expected: 3)	Answer
How many times does the letter ‘r’ appear in the word ‘strawberry’? Give only the numeric result.	3 · correct
In the word “strawberry”, how many ‘r’ characters are there? Give only the numeric result.	1 · wrong
Count the occurrences of the letter ‘r’ in ‘strawberry’. Give only the numeric result.	10 · wrong

Only the closing instruction changed

Counting ‘t’: “Give only the numeric result” → 1, “Answer with only the number” → 2.

Asked to copy 👩‍💻🚀🇧🇷

Qwen returned a *man* technologist and appended an unrelated Chinese sentence.

Input generation was not the bottleneck — oracles were

Conclusions

- Failures in 64.5% of replies, in every model and category
- Known answers turn differential and metamorphic signals into verdicts
- Paraphrases expose answers that depend on wording
- Coverage guidance covers the grammar, not more behaviors
- Report results under a content and a format criterion

Limitations

- Three small local models; hosted API not tested
- Only tasks with a computable answer are checked
- One seed in the main campaign; 3 in Ablation 2
- Manual judgments AI-assisted, pending review
- English, single-turn, templated prompts